

Shari's Digital ED Triage Model Pseudocode

Title: Pseudocode for a Digital Emergency Department Triage Risk Model

Overview

This digital triage model processes patient vital signs and basic clinical indicators to categorize patients into risk levels. The model uses cleaned patient data, checks for missing or abnormal values, applies clinical thresholds, calculates a risk score, and assigns a triage category. The goal is not to replace nurses or clinicians, but to support early identification of patients who may need urgent review.

Inputs

The model would use patient data such as:

Age
Gender
GCS
SBP
DBP
MAP
Pulse
Temperature
Respiratory Rate
FiO2
SpO2, if available
Pain score, if available

Risk Levels

The outputs are:

High Risk: Immediate clinical review needed
Moderate Risk: Needs timely assessment and monitoring
Low Risk: Stable but still requires routine triage assessment

Pseudocode for a Digital ED Triage Risk Model

START

1. Receive Patient Data

Input patient information:

- Age
- Gender
- GCS
- SBP
- DBP
- MAP
- Pulse
- Temperature
- Respiratory Rate
- FiO2
- SpO2, if available
- Pain score, if available
- Presenting complaint, if available

Initialize:

- risk_score = 0
- risk_level = "Unassigned"
- clinical_flags = empty list
- missing_data_flags = empty list
- critical_override = False

2. Check for Missing or Invalid Data

FOR each required vital sign:

IF value is missing:

- Add the vital sign name to missing_data_flags
- Add "Incomplete data: manual nurse review needed" to clinical_flags

IF value is impossible or non-numeric:

- Mark value as invalid
- Add "Invalid value detected" to clinical_flags

IF important data is missing:

Continue processing available data
Do not delete the patient record
Recommend manual nurse review

3. Prepare Blood Pressure Values

IF MAP is missing AND SBP and DBP are available:

Calculate MAP using:

$$\text{MAP} = (\text{SBP} + 2 \times \text{DBP}) / 3$$

IF MAP is abnormal but physiologically possible:

Preserve the value

Add "Abnormal MAP: clinical review needed" to clinical_flags

4. Check Consciousness Level Using GCS

IF GCS $\leq$ 8:

Add 3 points to risk_score

Add "Severely reduced consciousness" to clinical_flags

Set critical_override = True

ELSE IF GCS is between 9 and 12:

Add 2 points to risk_score

Add "Moderately reduced consciousness" to clinical_flags

ELSE IF GCS is between 13 and 14:

Add 1 point to risk_score

Add "Mildly reduced consciousness" to clinical_flags

ELSE IF GCS = 15:

Add 0 points

Add "Normal consciousness level" if needed

5. Check Blood Pressure and Perfusion

IF MAP $<$ 65:

Add 3 points to risk_score
Add "Low MAP / possible poor organ perfusion" to clinical_flags
Set critical_override = True

IF SBP < 90:

Add 3 points to risk_score
Add "Low SBP / possible hypotension or shock risk" to clinical_flags
Set critical_override = True

IF SBP > 180:

Add 2 points to risk_score
Add "Very high SBP / possible severe hypertension" to clinical_flags

6. Check Pulse

IF pulse < 50:

Add 2 points to risk_score
Add "Low pulse / possible bradycardia" to clinical_flags

ELSE IF pulse > 130:

Add 2 points to risk_score
Add "Very high pulse / urgent review needed" to clinical_flags

ELSE IF pulse > 100:

Add 1 point to risk_score
Add "Elevated pulse / possible tachycardia" to clinical_flags

7. Check Respiratory Rate

IF respiratory_rate < 8:

Add 3 points to risk_score
Add "Dangerously low respiratory rate" to clinical_flags
Set critical_override = True

ELSE IF respiratory_rate > 30:

Add 3 points to risk_score
Add "Very high respiratory rate / respiratory distress risk" to clinical_flags
Set critical_override = True

ELSE IF respiratory_rate > 20:

 Add 1 point to risk_score

 Add "Elevated respiratory rate" to clinical_flags

8. Check Temperature

IF temperature < 35:

 Add 2 points to risk_score

 Add "Hypothermia" to clinical_flags

ELSE IF temperature >= 39:

 Add 2 points to risk_score

 Add "High fever" to clinical_flags

ELSE IF temperature >= 38:

 Add 1 point to risk_score

 Add "Fever" to clinical_flags

9. Check Oxygenation and Oxygen Support

IF SpO2 is available:

 IF SpO2 < 90:

 Add 3 points to risk_score

 Add "Severe hypoxaemia" to clinical_flags

 Set critical_override = True

 ELSE IF SpO2 is between 90 and 94:

 Add 2 points to risk_score

 Add "Low oxygen saturation" to clinical_flags

IF FiO2 > 21:

 Add 1 point to risk_score

 Add "Patient may require oxygen support" to clinical_flags

10. Check for Multi-Sign Clinical Risk Patterns

IF MAP < 65 AND pulse > 100:

Add 2 additional points to risk_score

Add "Possible compensation or shock pattern: low MAP with high pulse" to clinical_flags

Set critical_override = True

IF respiratory_rate > 30 AND SpO2 < 94:

Add 2 additional points to risk_score

Add "Possible respiratory compromise pattern" to clinical_flags

Set critical_override = True

IF fever is present AND pulse > 100:

Add 1 additional point to risk_score

Add "Possible infection or systemic stress pattern" to clinical_flags

11. Assign Risk Level

IF critical_override = True:

risk_level = "High Risk"

ELSE IF risk_score >= 6:

risk_level = "High Risk"

ELSE IF risk_score is between 3 and 5:

risk_level = "Moderate Risk"

ELSE:

risk_level = "Low Risk"

12. Generate Explainable Triage Output

Display:

Patient risk level

Total risk score

Clinical flags

Missing data flags

Reason for risk category

Recommended triage action

IF risk_level = "High Risk":

Recommend immediate nurse or clinician review

ELSE IF risk_level = "Moderate Risk":

Recommend timely assessment and continued monitoring

ELSE IF risk_level = "Low Risk":

Recommend routine triage assessment

13. Human-in-the-Loop Safety Check

Send output to triage nurse.

Triage nurse reviews:

Vital signs

Clinical flags

Presenting complaint

Patient appearance

Clinical judgement

Final triage decision must be confirmed by the triage nurse.

END

Explanation

This digital triage model is designed to support the triage nurse. It works by taking the patient's vital signs and checking them against clinically meaningful ranges. Each abnormal finding adds to a risk score, but the system also has safety rules for serious warning signs. For example, a very low GCS, low MAP, low SBP, low oxygen saturation or dangerously abnormal respiratory rate would automatically flag the patient as high risk because those findings may suggest poor perfusion, reduced consciousness, shock or respiratory compromise. The model also looks for patterns, such as low MAP with a high pulse, which may suggest that the body is trying to compensate for poor circulation. Instead of only giving a final risk level, the system explains why the patient was placed in that category by listing the clinical flags and any missing or invalid data. This makes the output easier for nurses to review and trust. The final triage decision would still remain with the nurse, who would consider the patient's symptoms, appearance, history and overall clinical condition before deciding the next step.

Threshold Justification

Table 1:

Clinical Thresholds Used in the Digital ED Triage Model

Clinical Metric	Threshold Used in Model	Risk Level / Flag	Clinical Meaning and Triage Justification
GCS	≤ 8	Critical override / High Risk	Severely reduced consciousness. May indicate neurological compromise, airway risk, trauma, stroke, intoxication, or another serious condition requiring immediate review.
GCS	9–12	Moderate Risk	Moderate impairment in consciousness. The patient may still be unstable or deteriorating and should receive timely assessment.
GCS	13–14	Mild Risk Flag	Mild reduction in consciousness. This may still be important if it is new or linked to trauma, infection, or poor oxygenation.
MAP	< 65 mmHg	Critical override / High Risk	May indicate poor organ perfusion and possible shock. Low MAP can reduce blood flow to vital organs such as the brain, heart, and kidneys.
SBP	< 90 mmHg	Critical override / High Risk	May suggest hypotension, blood loss, dehydration, sepsis, or shock. This requires urgent review because circulation may be inadequate.
SBP	> 180 mmHg	Moderate/High Risk Flag	May suggest severe hypertension or cardiovascular strain, especially if symptoms such as chest pain, headache, or neurological signs are present.
Pulse	< 50 bpm	Moderate Risk Flag	May suggest bradycardia and can be concerning if linked to dizziness, hypotension, chest pain, altered mental status, or poor perfusion.

Pulse	> 100 bpm	Mild/Moderate Risk Flag	May suggest tachycardia caused by pain, fever, anxiety, dehydration, infection, bleeding, or shock.
Pulse	> 130 bpm	Higher Risk Flag	May indicate significant physiological stress and should prompt more urgent review, especially with low blood pressure, fever, or respiratory distress.
Respiratory Rate	< 8 breaths/min	Critical override / High Risk	May suggest respiratory depression, drug effect, neurological compromise, or severe deterioration.
Respiratory Rate	> 30 breaths/min	Critical override / High Risk	May suggest respiratory distress, sepsis, acidosis, hypoxia, or compensation for serious illness.
Respiratory Rate	> 20 breaths/min	Mild/Moderate Risk Flag	Indicates elevated respiratory effort and may be an early sign of infection, pain, anxiety, respiratory disease, or clinical deterioration.
Temperature	< 35°C	Moderate Risk Flag	May suggest hypothermia, exposure, sepsis, or severe illness and should be reviewed in context.
Temperature	≥ 38°C	Mild/Moderate Risk Flag	Suggests fever, which may be associated with infection, inflammation, or systemic illness.
Temperature	≥ 39°C	Higher Risk Flag	High fever may suggest more significant infection or systemic stress, especially with tachycardia or abnormal respiratory rate.
SpO ₂	< 90%	Critical override / High Risk	Suggests severe hypoxaemia and possible respiratory compromise, requiring urgent assessment or respiratory support.

SpO ₂	90–94%	Moderate Risk Flag	Indicates low oxygen saturation and should prompt assessment of breathing, oxygen needs, and respiratory or cardiac causes.
FiO ₂	> 21%	Oxygen Support Flag	Indicates the patient may be receiving supplemental oxygen. This is not automatically abnormal, but it suggests respiratory support or closer monitoring may be needed.
MAP + Pulse Pattern	MAP < 65 and Pulse > 100	Critical pattern / High Risk	May suggest compensation for poor circulation, possible shock, dehydration, bleeding, sepsis, or another acute stress state.
RR + SpO ₂ Pattern	RR > 30 and SpO ₂ < 94%	Critical pattern / High Risk	May suggest respiratory compromise, where the patient is breathing rapidly but still has low oxygen saturation.
Fever + Pulse Pattern	Temperature ≥ 38°C and Pulse > 100	Escalation Flag	May suggest infection or systemic stress and should increase concern when seen with other abnormal vital signs.

Note: The thresholds are used to support triage risk scoring and not to replace clinical judgment.

Limitations

This model is a simplified rule-based triage system and does not replace clinical judgement. Some patients may appear stable but still be critically ill, while others may have abnormal vital signs due to pain, anxiety, medication or chronic illness. The model should therefore be used as a decision-support tool, not as an automatic final decision-maker. It should also be tested, validated, and reviewed for bias before being used clinically.

Conclusion

In conclusion, this pseudocode brings together the main ideas I learned throughout Week 0: how triage data is collected, why vital signs matter, how abnormal values should be interpreted clinically, and why a digital system needs to be both structured and explainable. My model is

designed to process patient information, identify concerning vital signs, flag dangerous combinations such as low MAP with high pulse, and assign a risk level that can guide nurse review. I also included safety features such as missing-data flags, critical override rules, and human-in-the-loop confirmation because a triage model should support clinical judgement rather than replace it. Overall, this task helped me think like a designer of a clinical AI system by focusing not only on the final risk category but also on the reasoning and patient safety considerations behind that decision.

AI Use Statement:

AI tools were used to help structure and improve the clarity of this pseudocode and explanation. The final model design, clinical thresholds, and wording were reviewed and adapted to reflect my understanding of Week 0 triage concepts.